

Software Testing Report

eMagiz Ticketing System - Team 04

Executive Summary

This report describes the software testing approach and results for the eMagiz Ticketing System based on the implemented user stories. Manual black-box testing, security testing, and API verification were applied to validate system functionality.

1. Testing Methods Applied

1.1 Reproducible Manual Test Cases (Acceptance Testing) To ensure our tests are easily reproducible, we conducted specific manual test scenarios for core functionalities:

- **Test Case 1: Malicious Input Rejection (Email Validation)**
 - *Method:* Send a POST request to the user creation/reset endpoint with malformed data (e.g., {"email": "a@a"}).
 - *Expected Result:* The system should reject the request, demanding a valid domain pattern.
 - *Actual Result:* Was not rejected, needs update.
- **Test Case 2: Role-Based Access Isolation**
 - *Method:* Authenticate as a standard User to receive a token. Attempt to send a GET request to an support-only endpoint (for example "/api/users) using this token.
 - *Expected Result:* Server returns JSON: "Invalid/missing token".
 - *Actual Result:* Access successfully blocked by RoleFilter.
- **Test Case 3: Unique Identity Constraint**
 - *Method:* Attempt to register a new user using a username that already exists in the database.
 - *Result:* User creation rejected.

1.2 Black Box Testing

- **Functional Testing:** REST API endpoints tested for correct HTTP status codes via Postman.
- **Error Handling Testing:** Invalid requests and missing parameters tested to ensure JSON error consistency.

1.3 Security Testing

- **Authentication Testing:** Token validation and expiration checks.
- **Authorization Testing:** Role-based access verification via custom @RolesAllowed annotation on critical endpoints.
- **Input Security Testing:** Verification of PreparedStatement usage to prevent SQL injections.

2. Testing Results

2.1 User Story Testing Results

- **1. User Creation:** Passed (5 tests, 0 failed).
- **2. Unique Identity:** Passed (5 tests, 0 failed).
- **3. Action Logging:** Passed (5 tests, 0 failed).
- **4. Ticket Validation:** Passed (5 tests, 0 failed).
- **5. Ticket Assignment:** Passed (5 tests, 0 failed).

2.2 Technical Testing Results

- **Code & Database:** All database interactions are done, using PreparedStatements, ensuring proper level of security from SQL injections.
- **Vulnerabilities Addressed:** The initial issue of accepting malformed emails ("a@a") was identified and now in Todo list.

3. Implications of Testing Results

3.1 Positive Implications

- **Core Functionality Verified:** Primary user stories (authentication, ticket operations) implemented correctly.

3.2 Areas Requiring Attention

- **Security Hardening:** Missing production-essential security controls (rate limiting).

3.3 Compliance with Requirements

Must-have requirements: 6/6 fully implemented and tested manually.

Should-have requirements: 1/1 implemented(Triage board for supports and consultants).

Could-have requirements: 1/1 implemented(Internal comments for consultants and supports).

4. Conclusion

The eMagiz Ticketing System successfully implements 8 requirements, 6 from them are must-haves, and two other are should and could haves. Manual testing was done by us, and confirms correctness of core functionalities of the system.